

Most people are unaware of the simplicity of the internet design as it usually gets buried under the complex implementation details of various protocols. The paper aims to provide a digestible, easy to understand overview of the workings of the internet, focusing mainly on the "why" rather than "how". The paper dissects the various pieces of internet infrastructure and provides a high level overview of them. The author starts by describing the service model and discusses the three key mechanisms; routing, reliable delivery and name resolution in detail.

The author discusses that while ideally we would like to achieve all of the three, i.e. low latency, high bandwidth and reliable delivery, as internet is an amalgam of devices with various capabilities, packet networks support the lowest common denominator, i.e. best effort to deliver the packets but does not guarantee if, what or at what rate will the packets be delivered. The internet designers opted for this model in order to ensure minimal requirements on infrastructure which facilitated the rapid growth of the internet.

The paper then delves into the internet architecture, stating that modularity is a guiding principle. The functioning of the internet can be divided into two broad components, first being what the network infrastructure does and second being what the network support software on the host does. Network infrastructure can be further decomposed into three layers, with L1 handling most local tasks, L2 handling communication within the local network and L3 routing packets to the destination network. OS provides transport layer and other support to allow packets to reach the specific application. The author further provides insights on the functioning of each layer and the (inter and intra domain) routing mechanisms. Lastly, the author discusses the secrets to the internet's success and addresses the flaws and suboptimality of the internet design. These flaws include the lack of inherent security and lack of evolvability.

A limitation of this paper in my opinion is that it tries to summarize a lot of different concepts into a single article. While the author's aim is to, as the title suggests, extract the essential simplicity of the internet, to a novice reader this may feel like too much to swallow. Future work or extension of this paper could be to have separate articles that deal with the three major aspects of this paper. Keeping the same beginner friendly and simplistic tone, each article could further delve into network layers, routing mechanisms and the flaws respectively to keep things digestible.